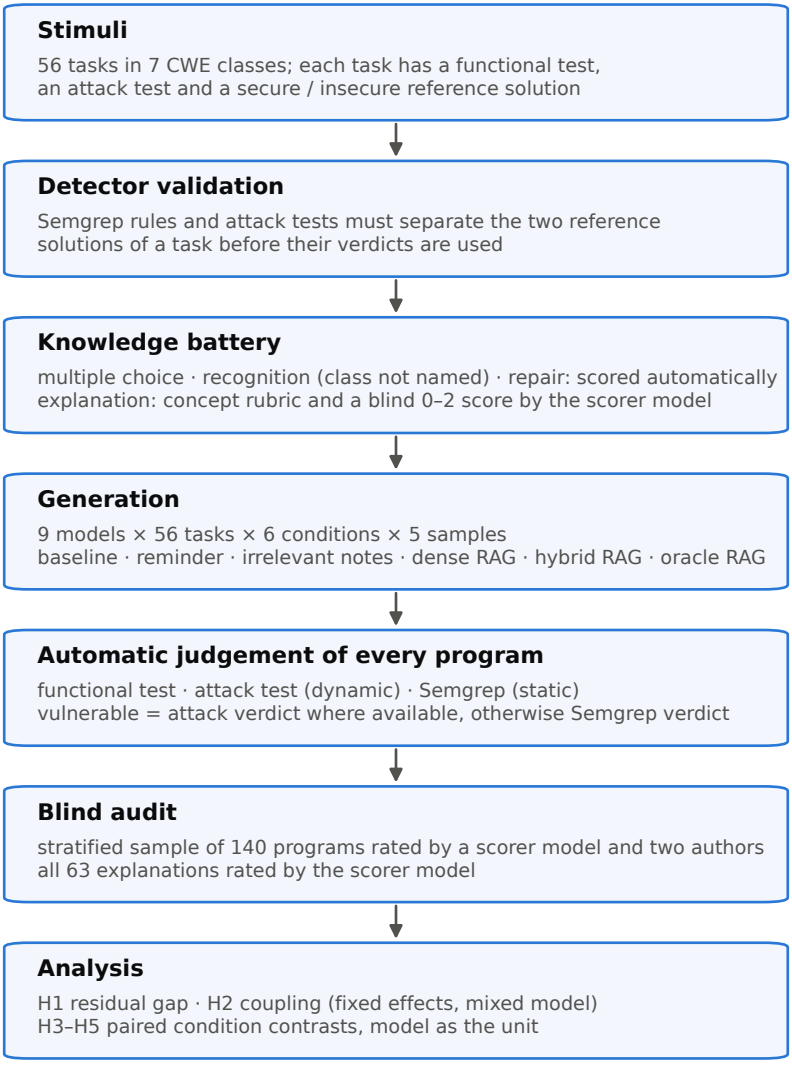
Stimuli
56 tasks in 7 CWE classes; each task has a functional test,
an attack test and a secure / insecure reference solution
Detector validation
Semgrep rules and attack tests must separate the two reference
solutions of a task before their verdicts are used
Knowledge battery
multiple choice · recognition (class not named) · repair: scored automatically
explanation: concept rubric and a blind 0–2 score by the scorer model
Generation
9 models × 56 tasks × 6 conditions × 5 samples
baseline · reminder · irrelevant notes · dense RAG · hybrid RAG · oracle RAG
Automatic judgement of every program
functional test · attack test (dynamic) · Semgrep (static)
vulnerable = attack verdict where available, otherwise Semgrep verdict
Blind audit
stratified sample of 140 programs rated by a scorer model and two authors
all 63 explanations rated by the scorer model
Analysis
H1 residual gap · H2 coupling (fixed effects, mixed model)
H3–H5 paired condition contrasts, model as the unit